

Individual Case #2
Ranking Air Hubs for AirCourier

BAN 120 : Analytics Case Studies

—

NAME : Vinchell Justin D'Costa
STUDENT ID : 133422220

Executive Summary

AirCourier, a startup company aiming to offer efficient letter, document, and package delivery services across Canada, seeks to establish strategic hubs to optimize logistics and reduce delivery times. This report analyzes climate data from various Canadian cities to recommend the most favorable locations for these hubs, prioritizing year-round operations without interruptions.

Introduction

AirCourier, a dynamic startup company, envisions revolutionizing the delivery landscape across Canada by offering swift and efficient services for letters, documents, and packages. To achieve this goal, AirCourier aims to establish strategically located hubs that can seamlessly receive and reroute shipments, streamlining logistics and significantly reducing delivery times. The pivotal criterion for selecting these hub locations is to ensure a consistently dry environment all year round, eliminating potential operational interruptions.

In pursuit of this mission, AirCourier is considering several cities as potential hub locations. The list comprises prominent Canadian cities known for their economic significance and regional connectivity. Among the contenders are Calgary, Edmonton, Vancouver, Winnipeg, Fredericton, St. John's, Halifax, Toronto, Charlottetown, Montréal, Québec City, and Saskatoon.

To make well-informed decisions, AirCourier has meticulously gathered and analyzed climate data for these cities from credible sources such as Climate Canada. The data spans the years 2021 to 2050, based on the SSP 4.5 moderate emissions scenario. Specifically, the focus is on key precipitation metrics, such as average total precipitation and average number of rainy days, to assess and rank the cities in terms of their suitability for AirCourier's hubs.

This report aims to provide a comprehensive analysis of the gathered data, presenting a ranked list of the most desirable cities to serve as AirCourier's hub locations. The findings will enable AirCourier to make strategic choices that align with its vision of providing seamless and efficient delivery services across Canada.

Findings

Wet Days data (2021 to 2050)

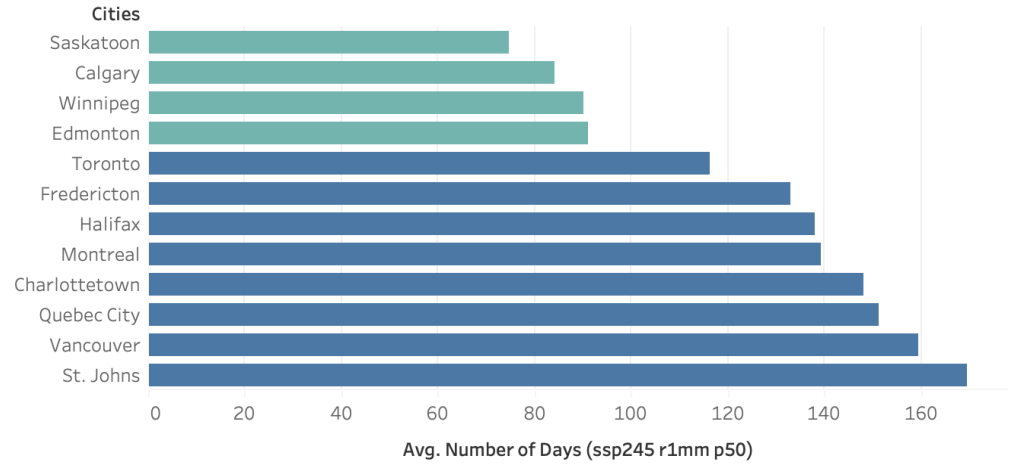
Map Showing Average Number of Wet Days (2021-2050)



Map based on Longitude and Latitude. Color shows average of Number of Days (ssp245 r1mm p50). The marks are labeled by Cities. The data is filtered on Time Year, which keeps 2021, 2031, 2041 and 2051.

Avg. Number of Days (s..
74.52 169.49

Bar Graph Showing Average Number of Wet Days (2021-2050)

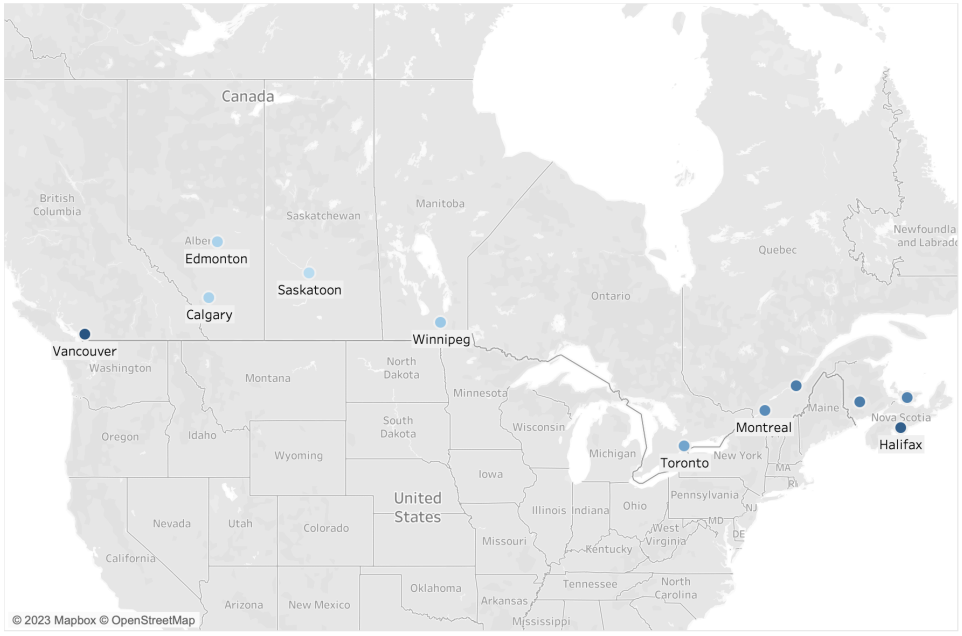


Average of Number of Days (ssp245 r1mm p50) for each Cities. Color shows details about LeastWetDays. The data is filtered on Time Year, which keeps 2021, 2031, 2041 and 2051.

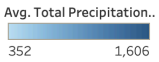
LeastWetDays
Least Wet Days
Other

Total Precipitation data (2021 to 2050)

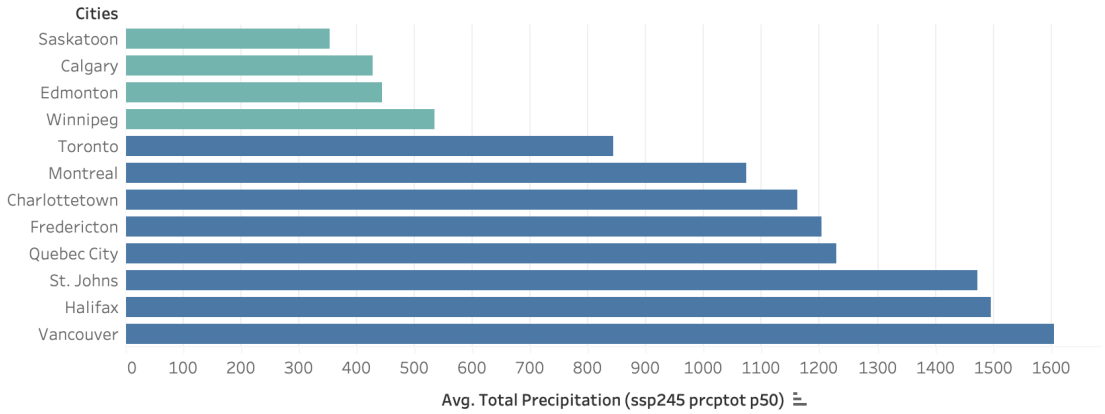
Map Showing Average Total Precipitation (2021-2050)



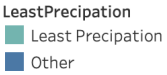
Map based on Longitude and Latitude. Color shows average of Total Precipitation (ssp245 prcptot p50). The marks are labeled by Cities. The data is filtered on Time Year, which keeps 2021, 2031, 2041 and 2051.



Bar Graph Showing Average Total Precipitation (2021-2050)



Average of Total Precipitation (ssp245 prcptot p50) for each Cities. Color shows details about LeastPrecipation. The data is filtered on Time Year, which keeps 2021, 2031, 2041 and 2051.



Ranking Results

1. **Saskatoon:** With the least average wet days (75) and total precipitation (352 mm), Saskatoon emerges as the most favorable location for AirCourier's hub due to its relatively dry climate, making it ideal for year-round operations.

2. **Calgary:** Securing the second position, Calgary experiences 84 average wet days and 427 mm of total precipitation, offering a conducive climate for AirCourier's hub operations with fewer interruptions.

3. **Edmonton:** Edmonton follows closely, with 91 average wet days and 443 mm of total precipitation, making it a viable choice for establishing a hub with manageable precipitation levels.

4. **Winnipeg:** Ranking fourth, Winnipeg records 90 average wet days and 534 mm of total precipitation, providing favorable conditions for operating a hub throughout the year.

5. **Toronto:** Toronto takes the fifth spot, boasting 116 average wet days and 844 mm of total precipitation, making it a suitable location for AirCourier's hub with moderately manageable precipitation.

6. **Montréal:** In sixth place, Montréal experiences 139 average wet days and 1073 mm of total precipitation, providing acceptable conditions for a hub, albeit with higher precipitation levels.

7. **Charlottetown:** Ranking seventh, Charlottetown has 148 average wet days and 1163 mm of total precipitation, making it a less favorable choice due to higher precipitation levels.

8. **Québec City:** In the eighth position, Québec City receives 151 average wet days and 1228 mm of total precipitation, offering less favorable conditions for year-round operations.

9. **Fredericton:** Securing the ninth spot, Fredericton records 133 average wet days and 1203 mm of total precipitation, making it a less suitable option for AirCourier's hub.

10. **Halifax:** In tenth place, Halifax experiences 138 average wet days and 1496 mm of total precipitation, presenting less favorable conditions for hub operations.

11. **St. John's:** Ranking eleventh, St. John's has 169 average wet days and 1474 mm of total precipitation, making it one of the least favorable choices due to its higher precipitation levels.

12. **Vancouver:** Securing the twelfth position, Vancouver experiences the highest average wet days (159) and total precipitation (1606 mm), making it the least suitable location for AirCourier's hub with its consistently wet climate.

Recommendations

Based on the detailed climate data analysis, the following three locations are recommended as the most suitable hubs for AirCourier, considering their dry climate and manageable precipitation levels:

I. Saskatoon

With an average of only 75 wet days and a total precipitation of 352 mm annually, Saskatoon emerges as the top choice. Its remarkably low precipitation ensures a dry environment for year-round operations, minimizing potential delays and interruptions. Additionally, the relatively mild precipitation levels make it an optimal location for efficiently handling AirCourier's deliveries.

II. Calgary

Securing the second position, Calgary offers an appealing climate with 84 average wet days and a total precipitation of 427 mm annually. The city's relatively dry conditions create a conducive environment for hub operations, enabling consistent and smooth delivery services. The manageable precipitation levels further reinforce its suitability as a hub location for AirCourier.

III. Edmonton

Following closely behind Calgary, Edmonton experiences an average of 91 wet days and a total precipitation of 443 mm per year. This places it as a viable choice for establishing a hub, as its precipitation levels remain within acceptable limits. The relatively dry climate ensures operational continuity, while the slightly higher precipitation compared to Saskatoon and Calgary can still be effectively managed by AirCourier.

IV. Toronto

Although experiencing 116 average wet days and 844 mm of total precipitation, Toronto ranks fifth and presents a suitable location for AirCourier's hub. As a hub on the east side of Canada, it complements the other three hubs located on the west side, enhancing regional connectivity and providing a strategic advantage.

These four recommended locations offer a balanced mix of drier climates, fewer wet days, and moderate precipitation. By establishing hubs in these cities, AirCourier can optimize its logistics, reduce delivery times, and provide reliable services to customers across Canada.

Conclusion

In conclusion, the analysis of climate data for various Canadian cities has provided valuable insights for AirCourier in selecting strategic hub locations. Saskatoon, Calgary, Edmonton and Toronto emerge as the top three recommendations due to their dry climate and manageable precipitation levels. These locations offer the ideal conditions for year-round operations, ensuring efficient and reliable delivery services. By establishing hubs in these cities, AirCourier can optimize its logistics and achieve its vision of transforming the delivery landscape across Canada.

Link to Presentation

: https://express.adobe.com/page/vIpfx0HzTOmvi/